

Unsteady Simulation — Performance & Inputs

Overall performance

Apogee (AGL)	Apogee (ASL)	Burn time	Total impulse
6,860.3 m	7,160.3 m	15.19 s	38,856 N·s
Peak thrust	Average thrust	Pad T/W	Peak chamber p
5.42 kN	2.56 kN	8.31	44.0 bar
Peak chamber T	Average O/F	Oxidizer used	Fuel used
3,025 K	4.01	14.36 kg	2.99 kg
Oxidizer left	Fuel left	Sim wall clock	Total timesteps
1.14 kg	3.01 kg	95.39 s	2,797

Per-phase performance

Phase	Dur [s]	Impulse [Ns]	Peak F [kN]	Avg O/F	Peak Pc [bar]
1: Ignition	0.00	0	0.20	6.26	6.0
2: Liquid blowdown	8.22	37,730	5.42	5.36	44.0
3: Gaseous blowdown	6.97	1,126	3.61	2.35	30.0
4: 5: Coast	20.18	peak 253.7 m/s, term 25.0			
6: Drogue descent	60.00	peak 64.8 m/s, term 57.0			

Rocket inputs

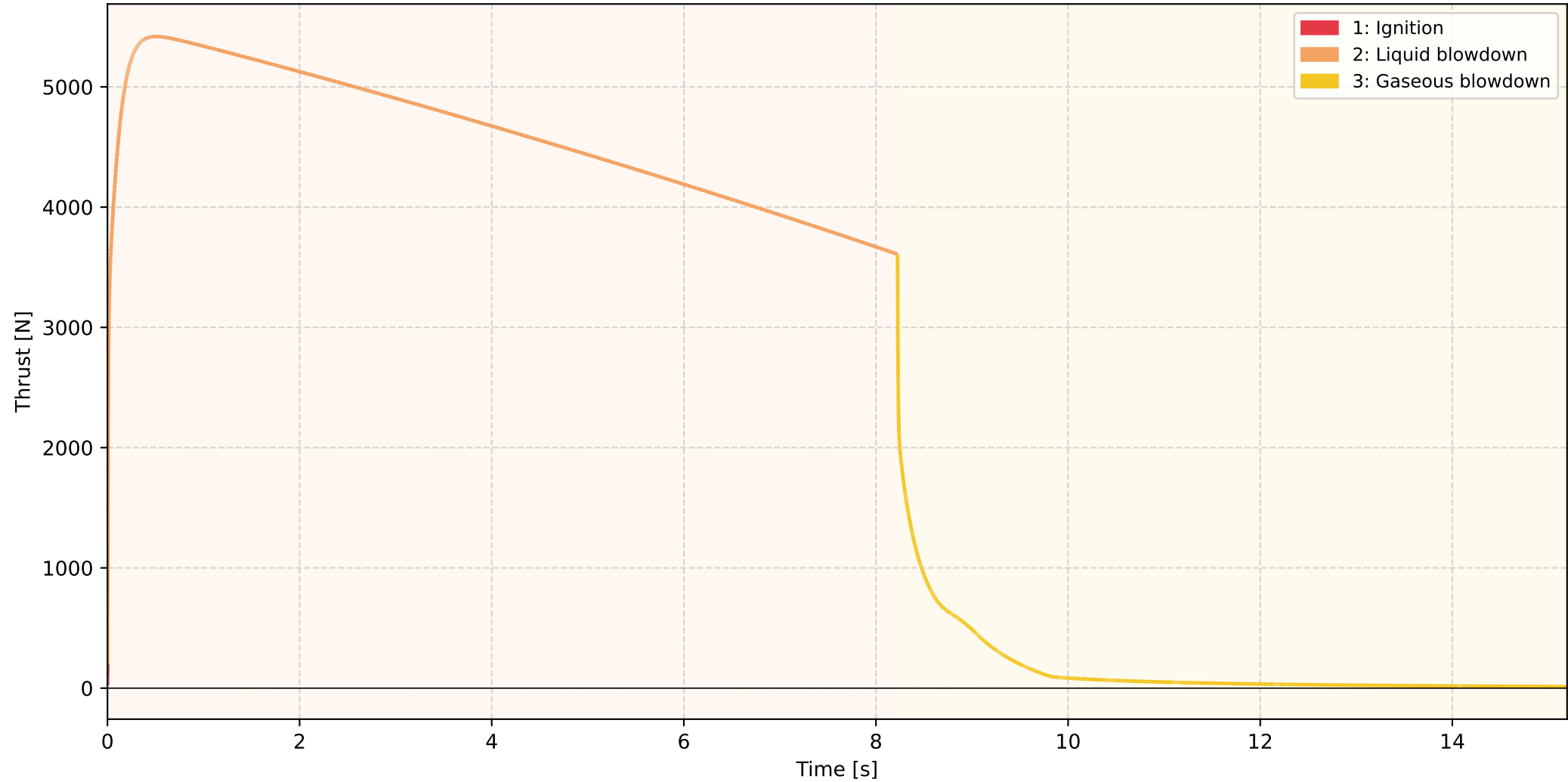
Launch site	
Altitude ASL	300.0 m
Tank (CV1)	
Internal radius	0.1000 m
Shell length	0.500 m
Volume	0.01570 m³
Initial temp	298.15 K
Ox mass loaded	15.500 kg
Ullage fraction	0.100
Dip tube OD/2	0.00600 m
Dip tube ID/2	0.00400 m
Dip tube length	0.450 m
Valve (CV2)	
Sigmoid t½	0.050 s
Sigmoid k	10.00
Injector (CV3)	
Cd	0.800
Number holes	36
Hole area	0.00000180 m²
Feed Δp	300,000 Pa
Chamber (CV4)	
Fuel length	0.450 m
Fuel density	900.0 kg/m³
Fuel OR	0.1000 m
Fuel mass loaded	6.000 kg
Regression a	0.0001320
Regression n	0.555
Pre-chamber V	0.001000 m³
Post-chamber V	0.001500 m³
Nozzle (CV5)	
Throat radius	0.0160 m
Exit radius	0.0450 m
Trajectory (CV6)	
Dry mass	45.00 kg
Cd	0.640
Frontal area	0.03200 m²
Launch angle	6.0 °
Drogue Cd	1.20
Drogue area	0.28 m²
Main deploy AGL	450.0 m
Main Cd	1.50
Main area	9.62 m²

Event Log & Warnings		
Event log (4 events)		
t [s]	Type	Message
0.001	PHASE_TRANSITION	Event 'event_ignition_pressure_reached' triggered. Exiting phase_1 to phase_2
8.225	PHASE_TRANSITION	Event 'event_liquid_depleted' triggered. Exiting phase_2 to phase_3
15.194	PHASE_TRANSITION	Event 'event_chamber_near_ambient_during_gaseous_blowdown' triggered. Exiting phase_3 to phase_5
35.388	PHASE_TRANSITION	Event 'event_apogee_reached' triggered. Exiting phase_5 to phase_6

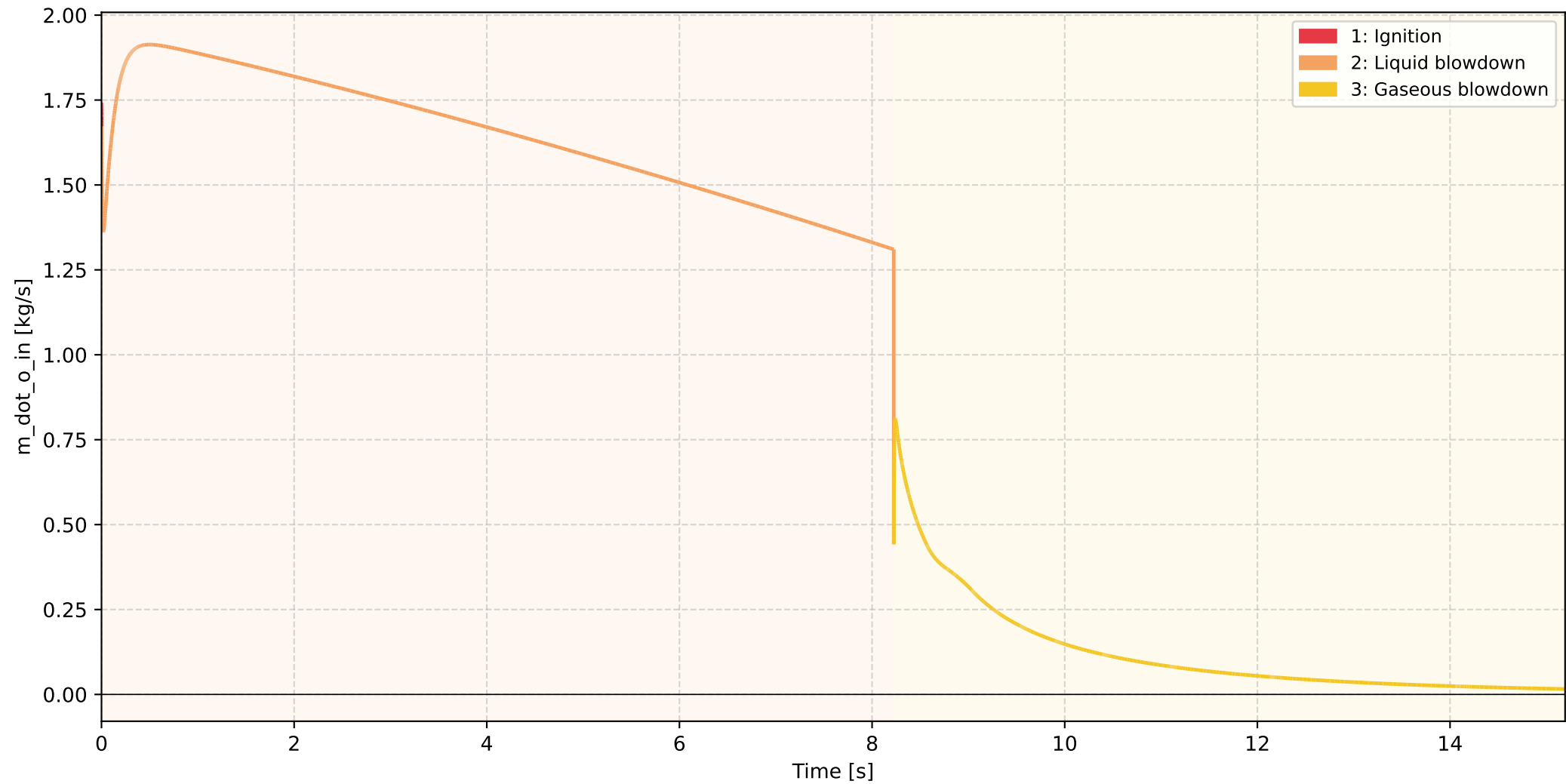
Warnings — overall level: advisory

Warning	Severity	Occurrences	Message
CEA_outside_table_bounds	debug	291	One or more NASA-CEA inputs are outside the precomputed table's range
CEA_outside_tested_range	advisory	62	One or more NASA-CEA inputs are outside the range the program was designed to handle

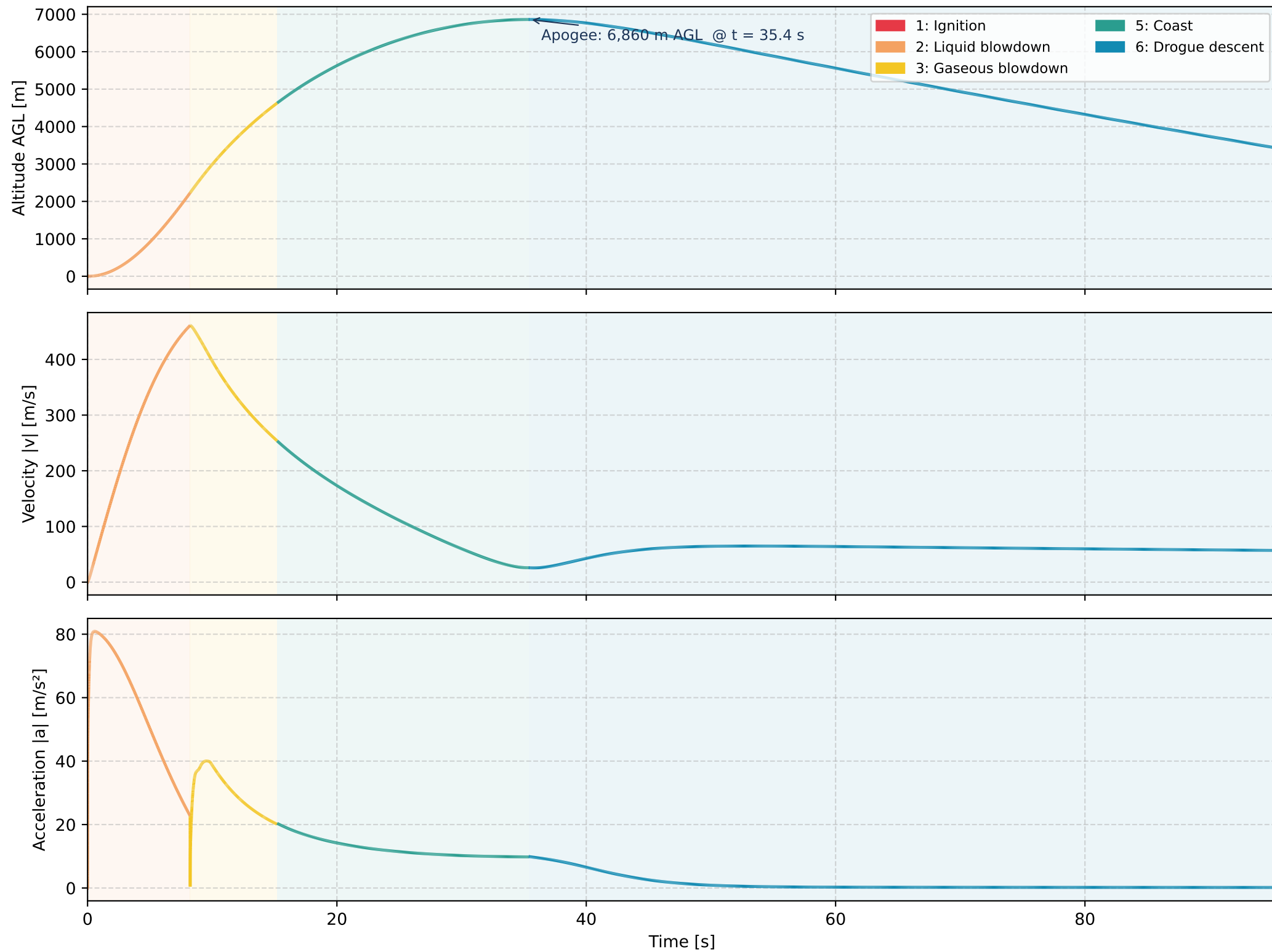
Thrust vs. time (burn only)



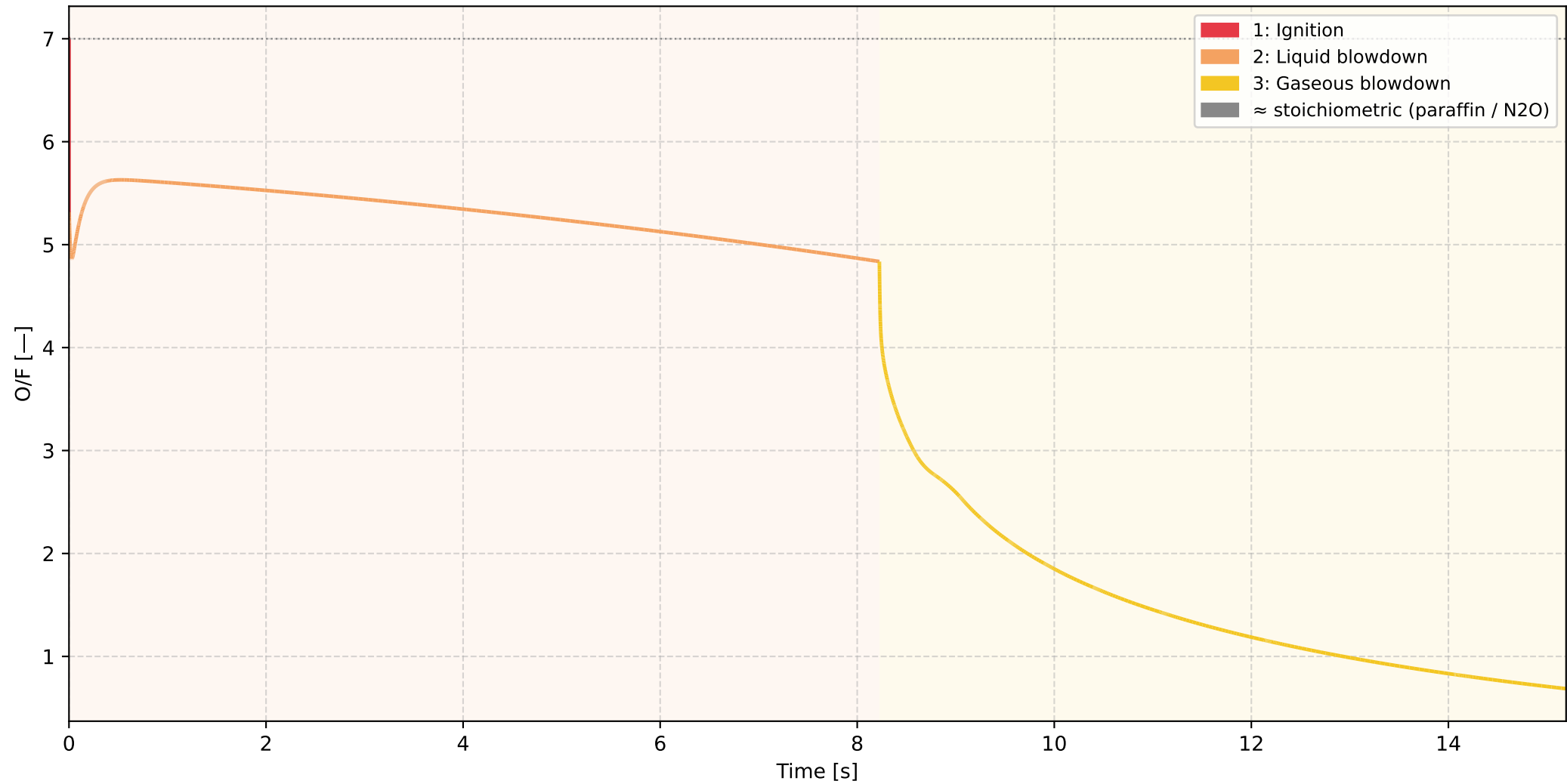
Injector mass flow vs. time (burn only)



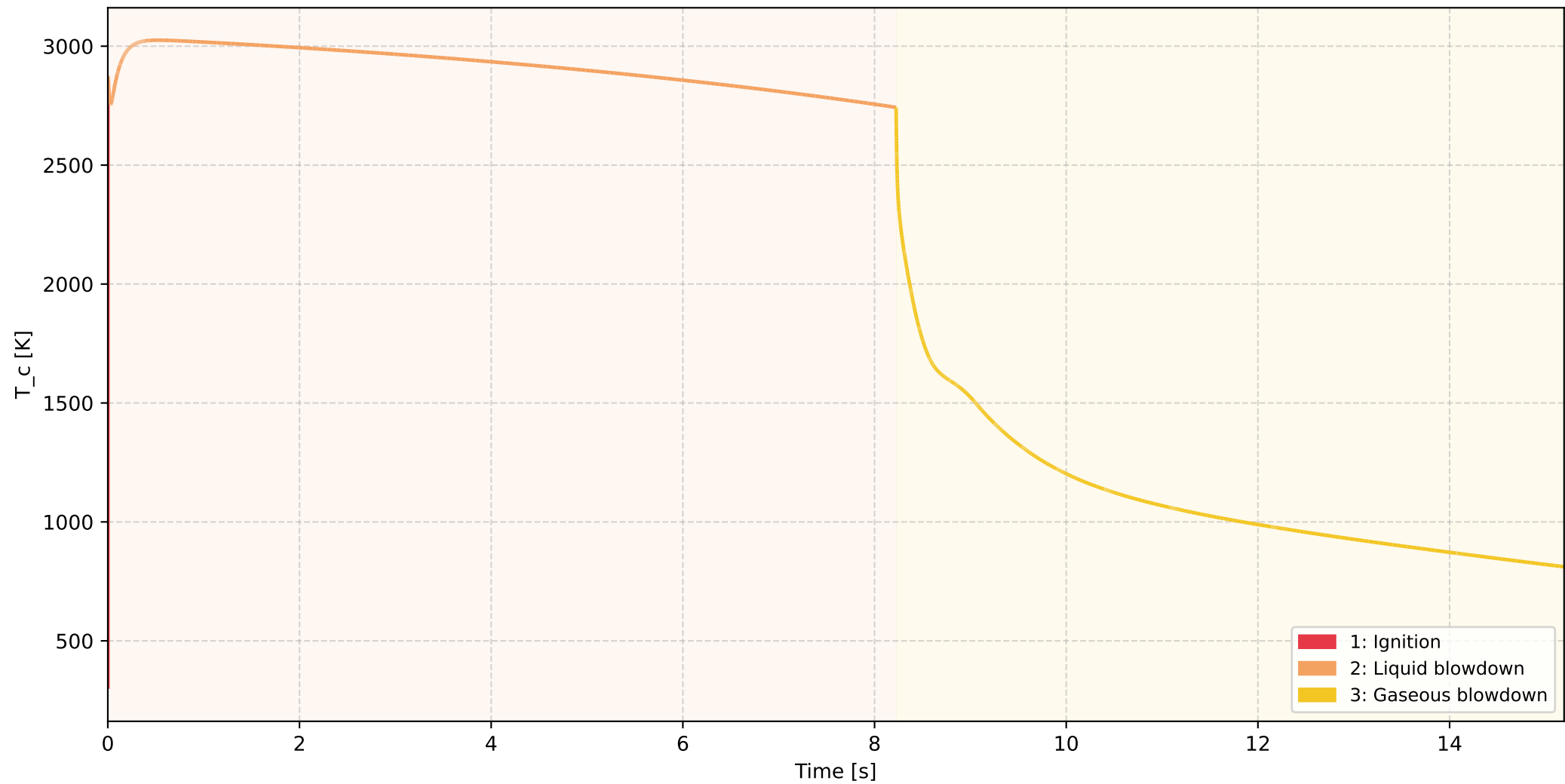
Rocket kinematics



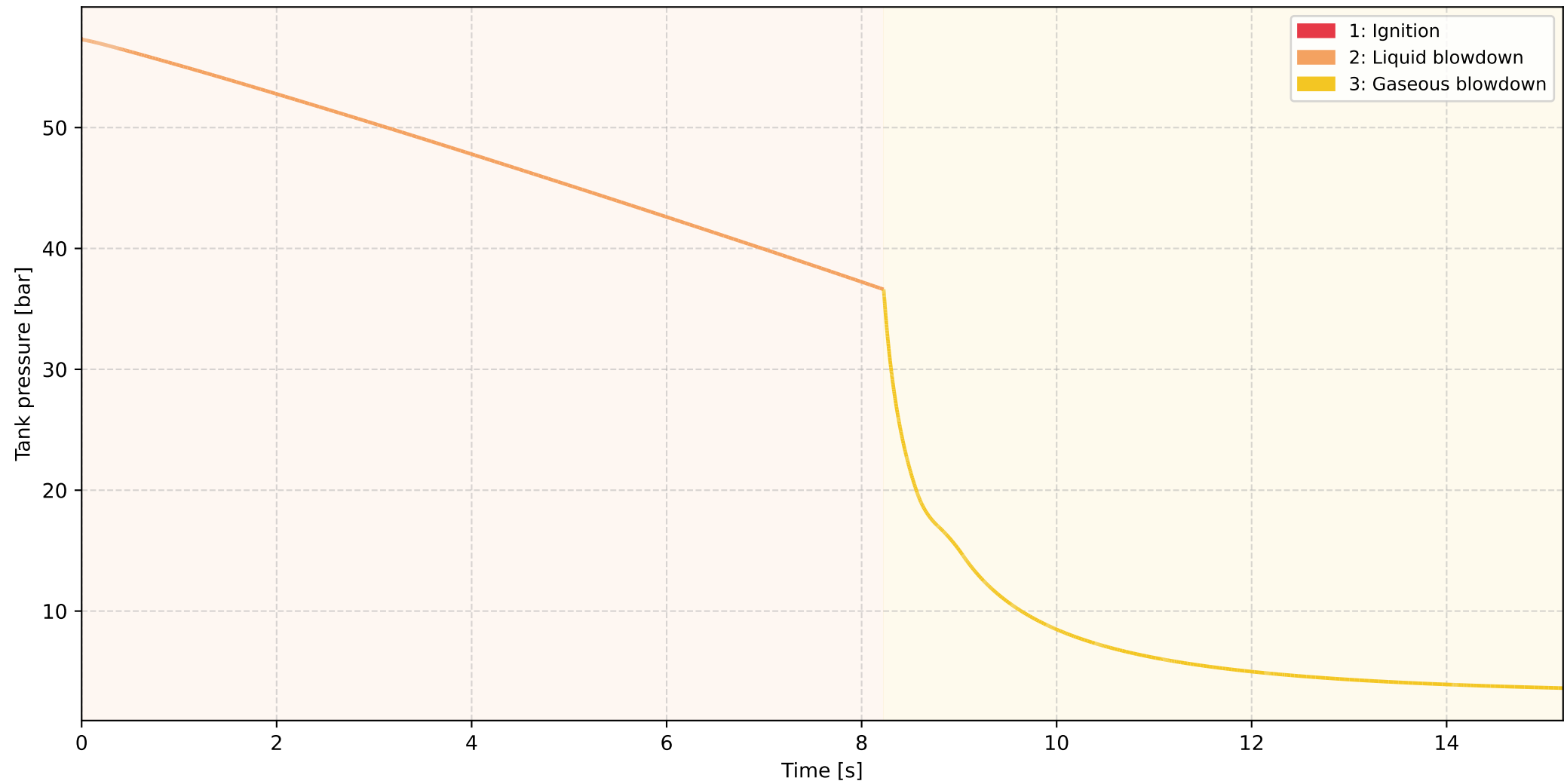
O/F ratio vs. time (burn only)



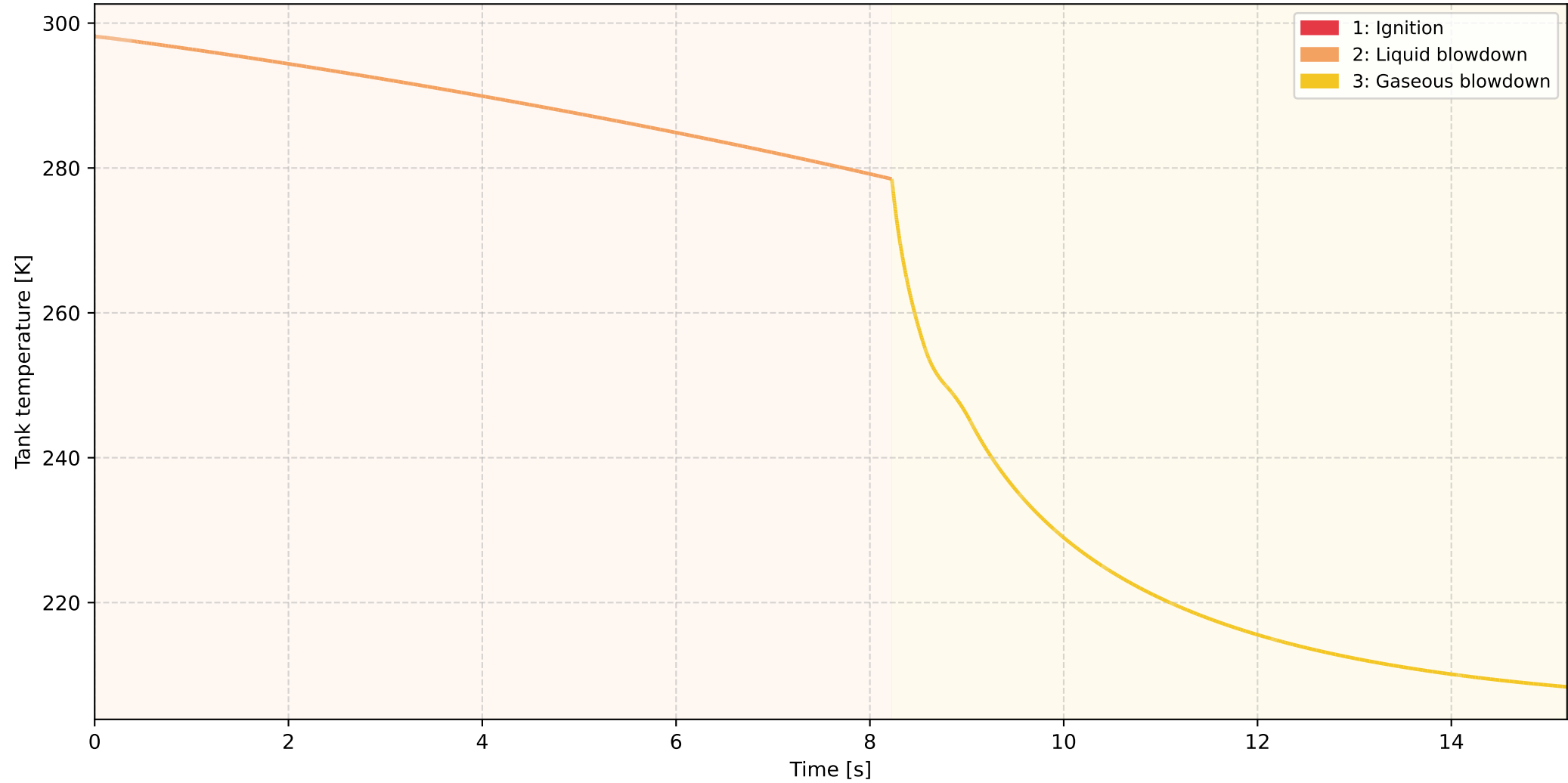
Chamber temperature vs. time (burn only)



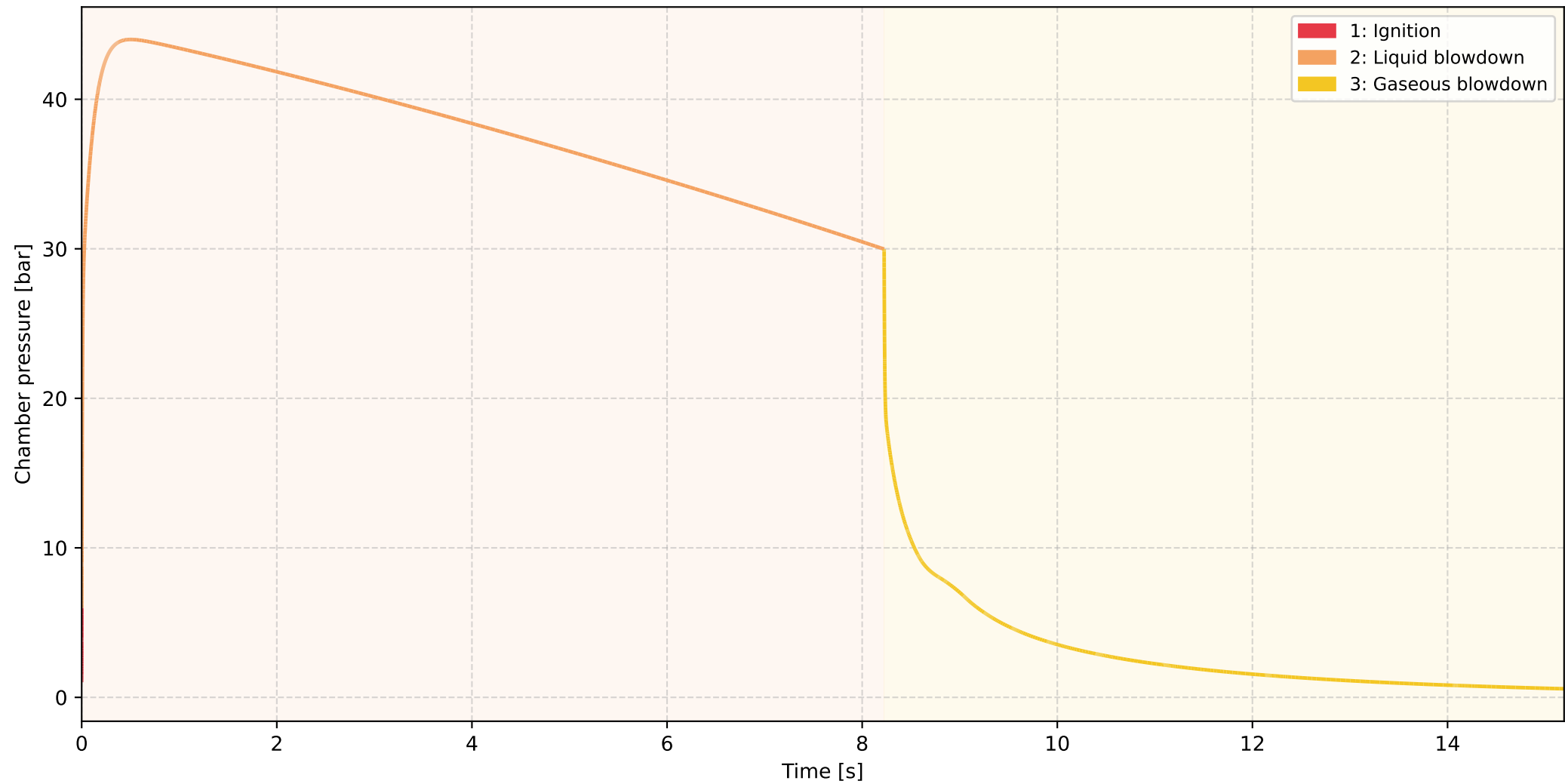
Tank pressure vs. time (burn only)



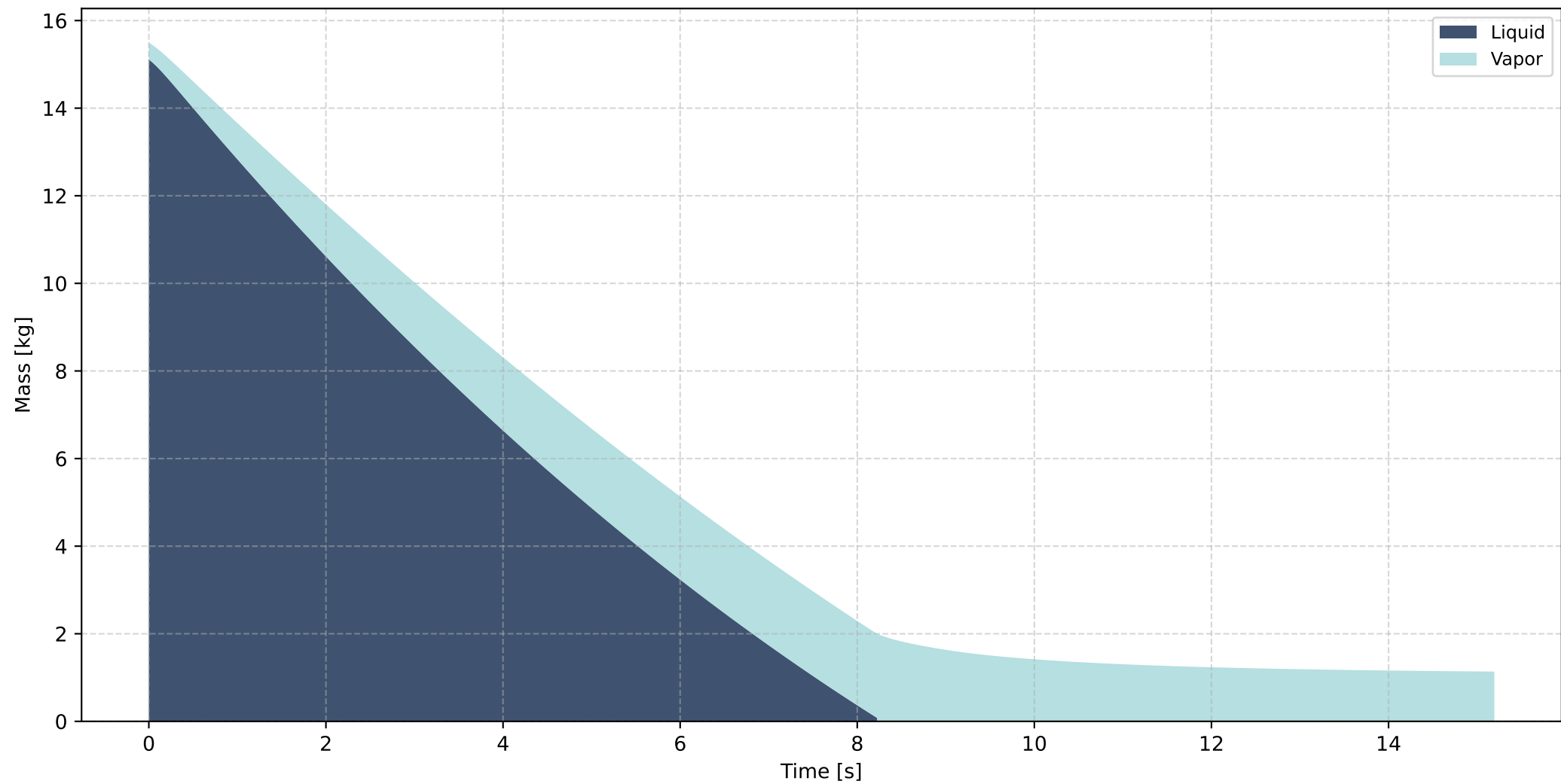
Tank temperature vs. time (burn only)



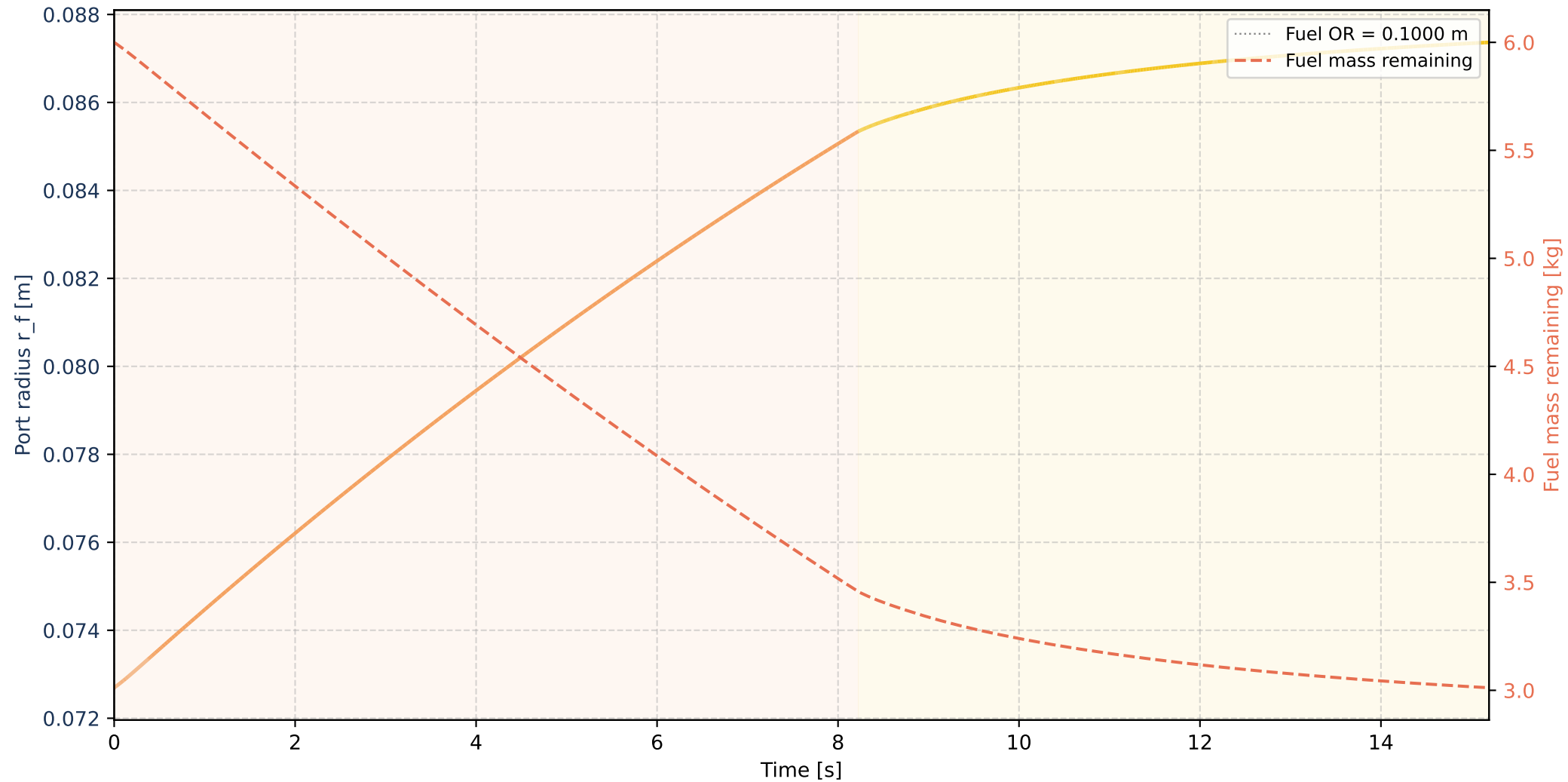
Chamber pressure vs. time (burn only)



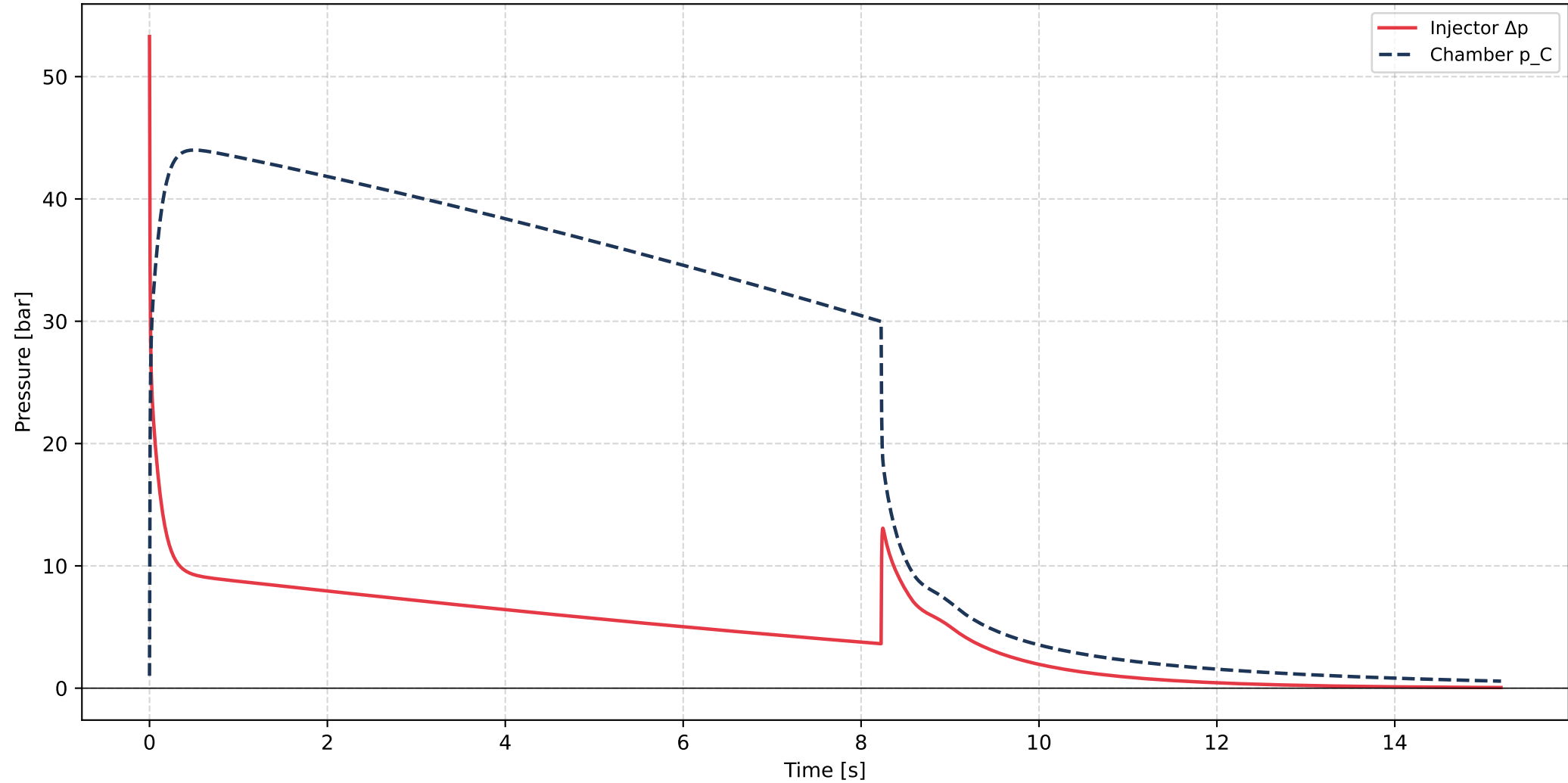
Oxidizer remaining vs. time (burn only)



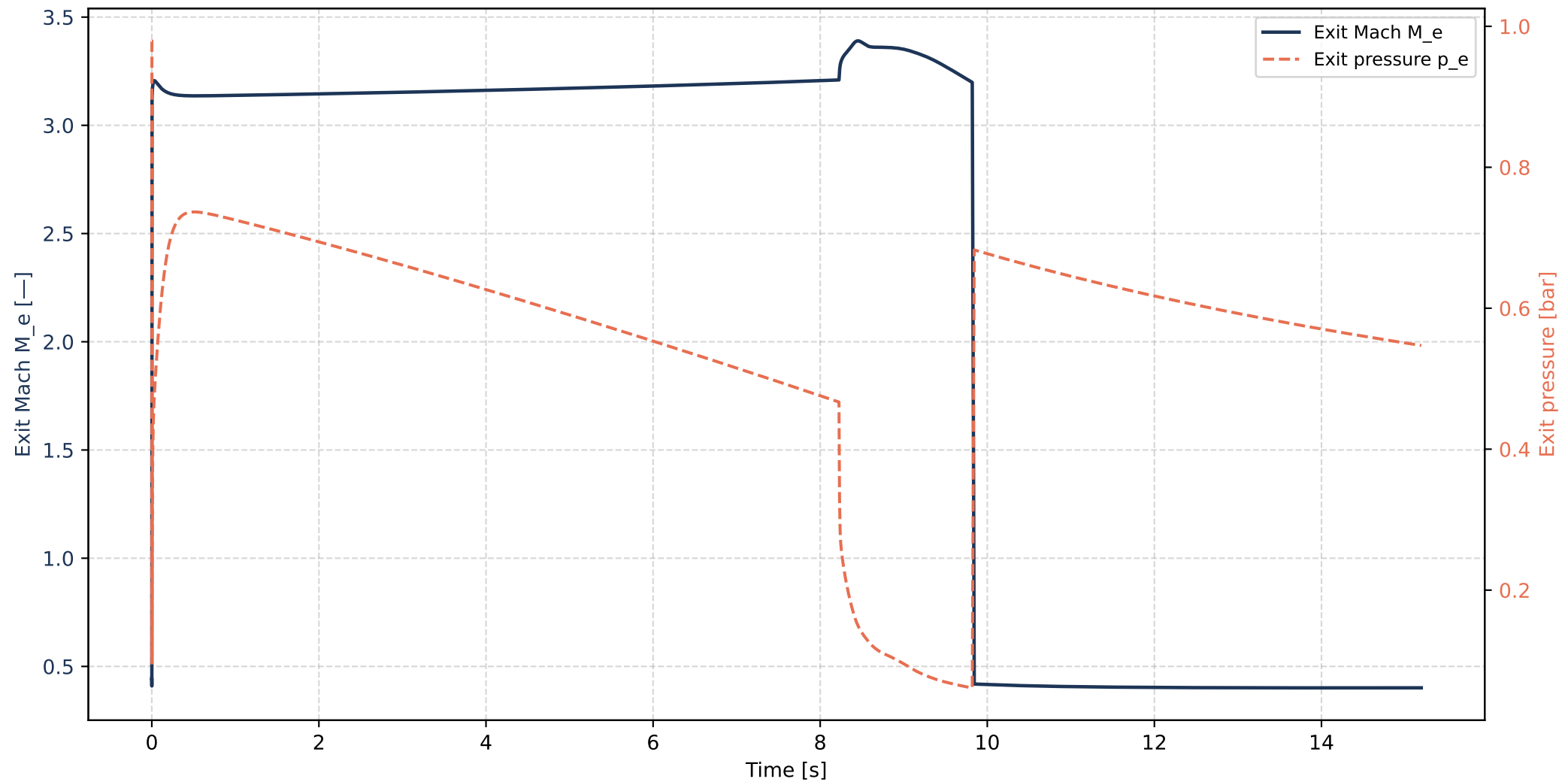
Fuel grain state vs. time (burn only)



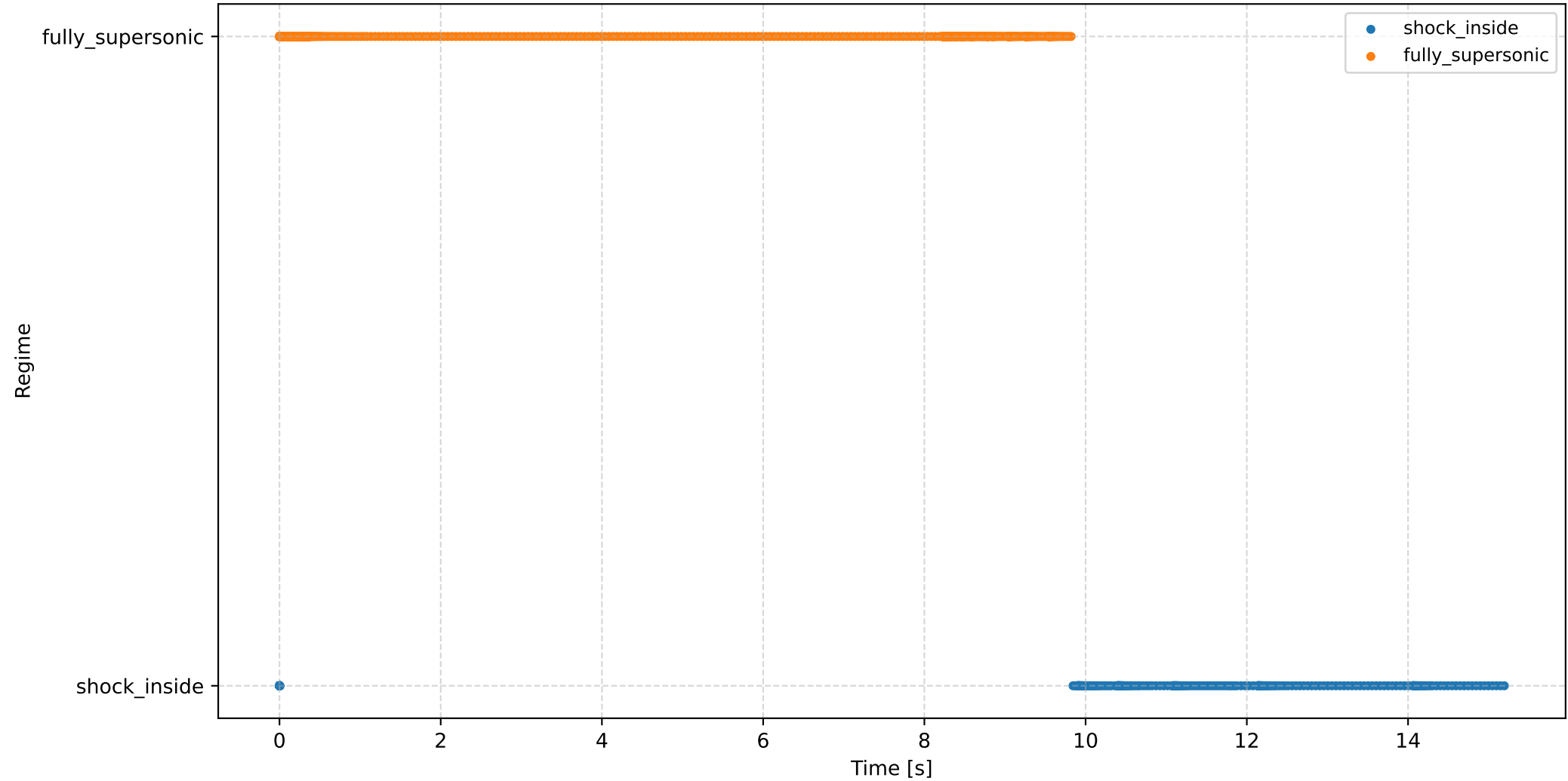
Injector Δp & chamber pressure vs. time (burn only)



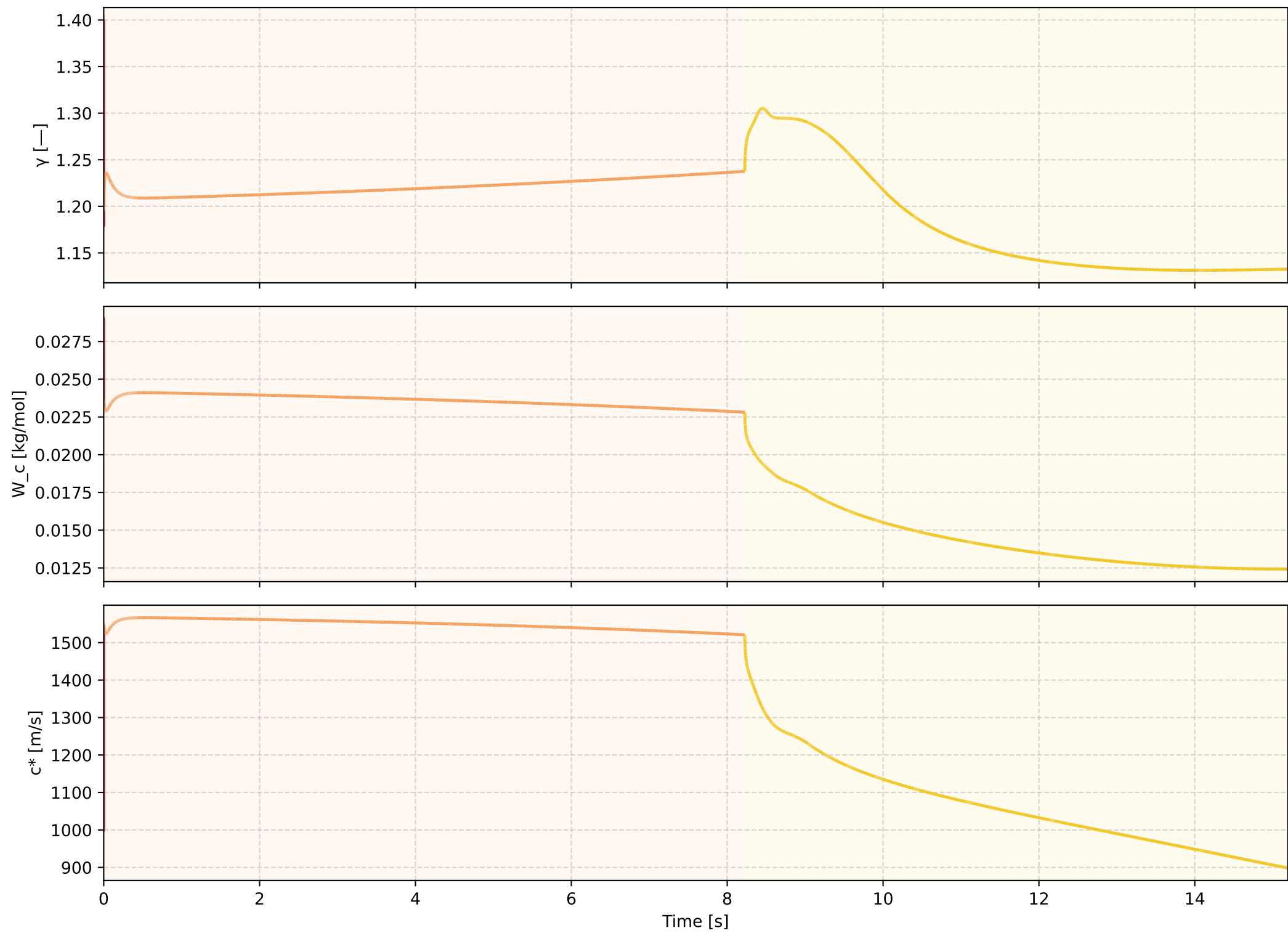
Nozzle exit conditions vs. time (burn only)



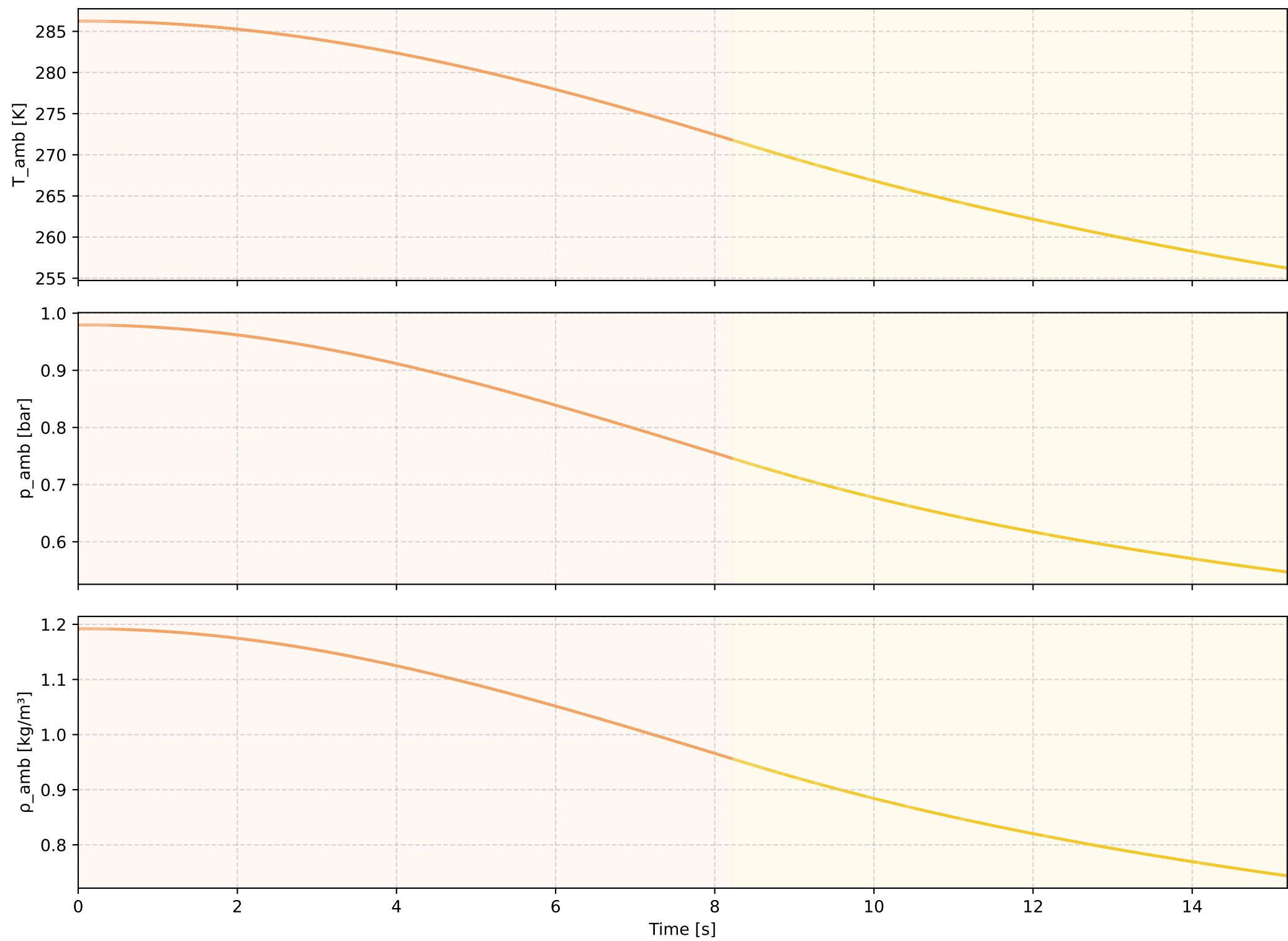
Nozzle flow regime vs. time (burn only)



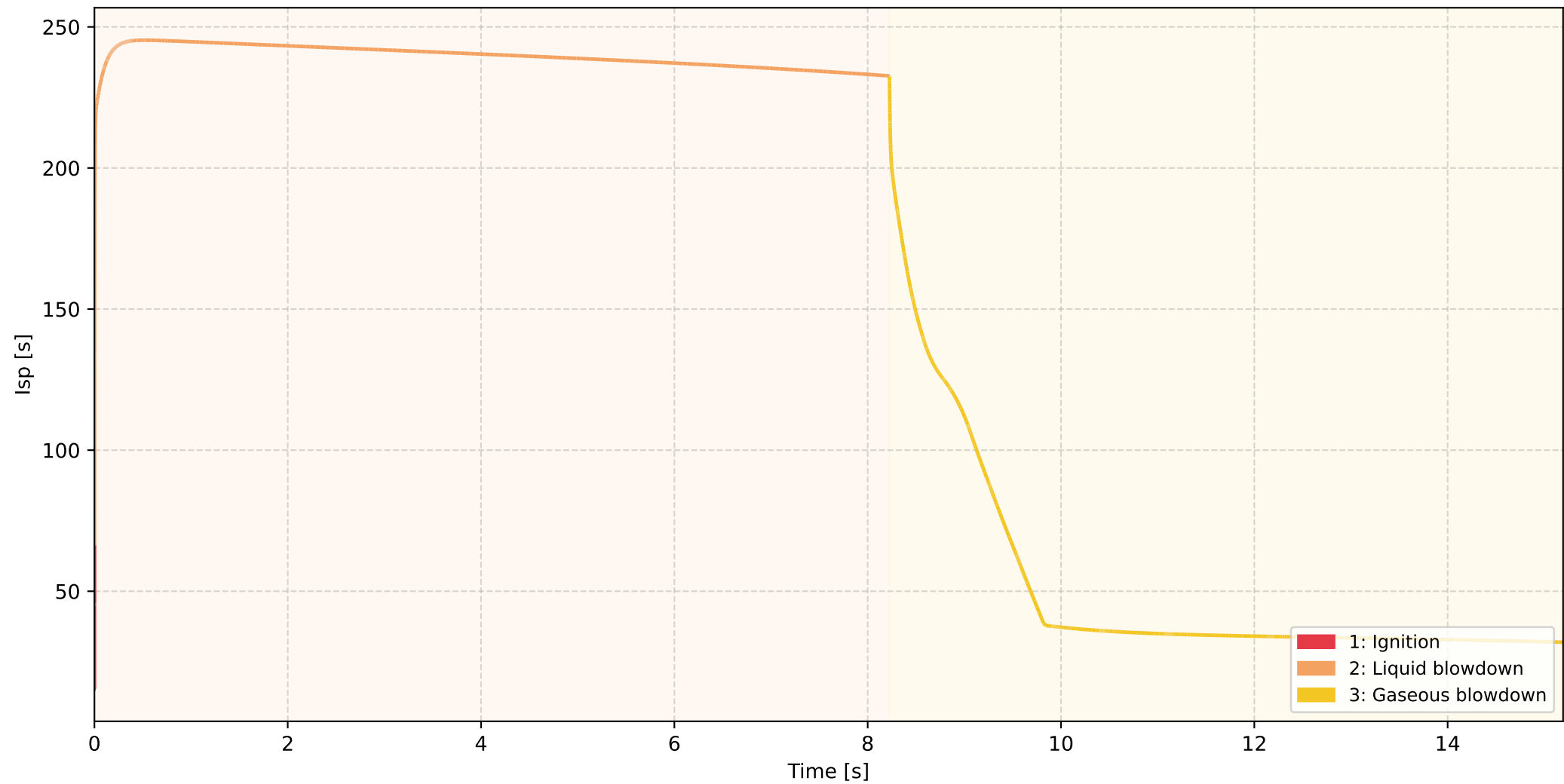
Combustion properties vs. time (burn only)



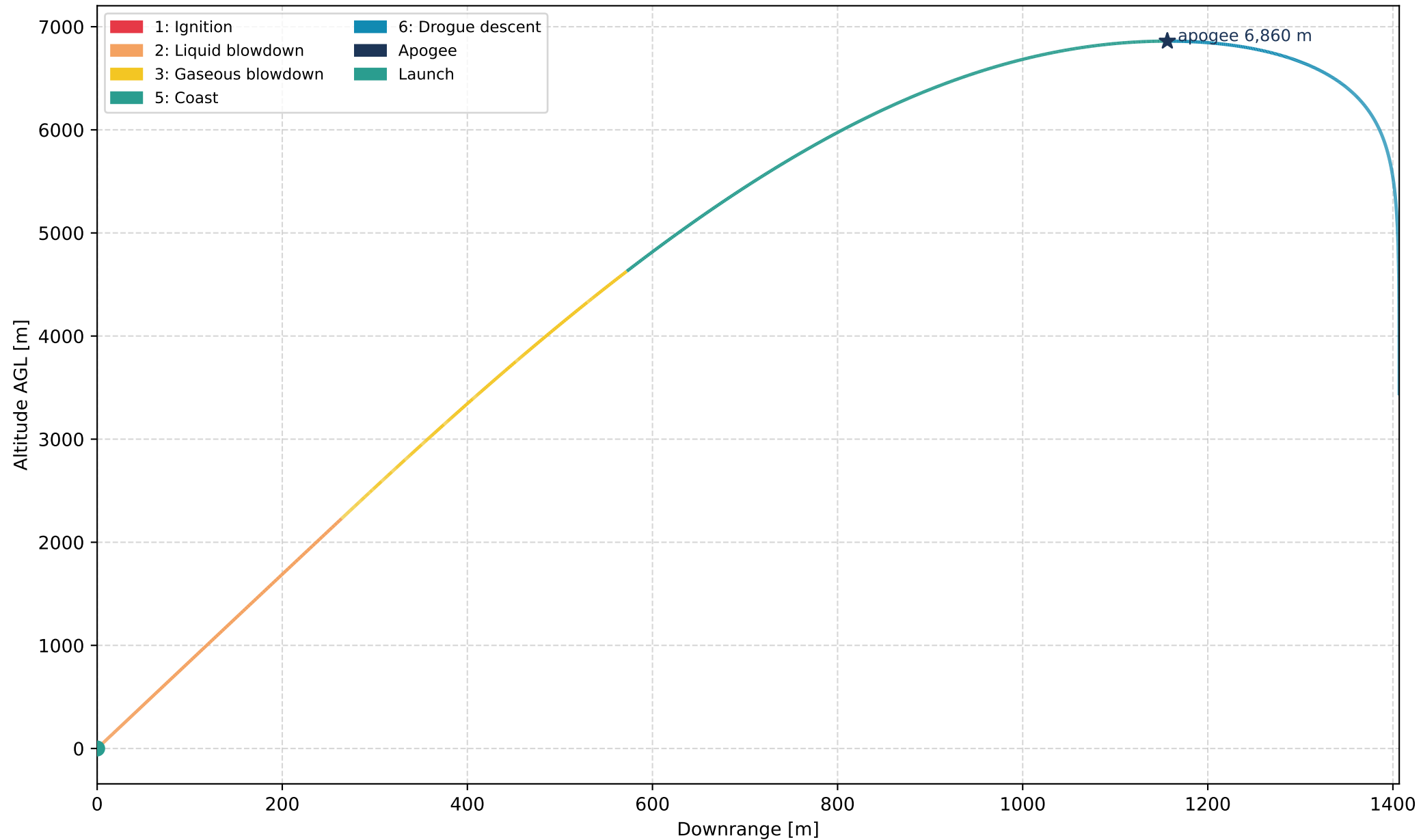
Ambient atmosphere vs. time (burn only)



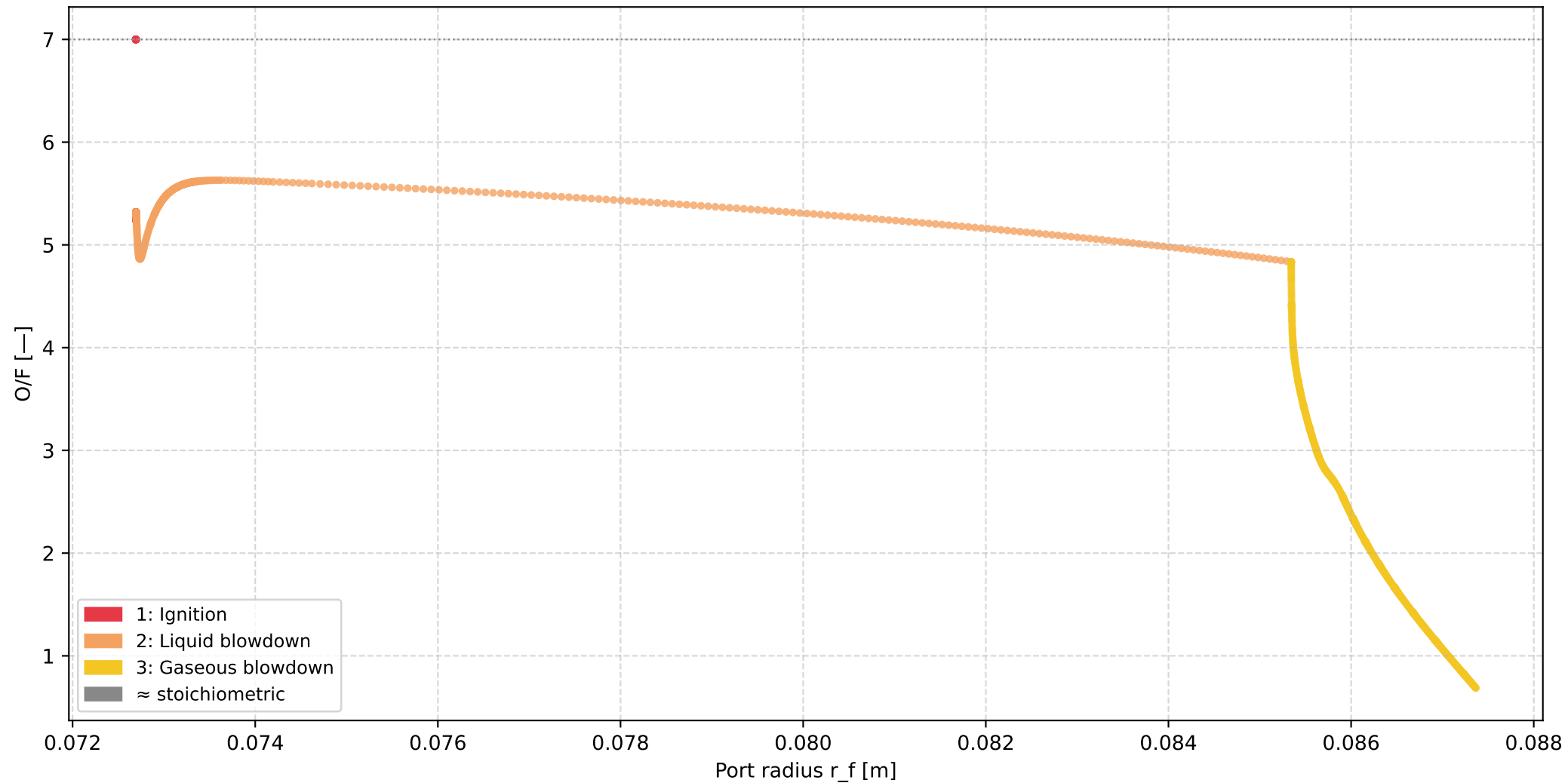
Specific impulse vs. time (burn only)



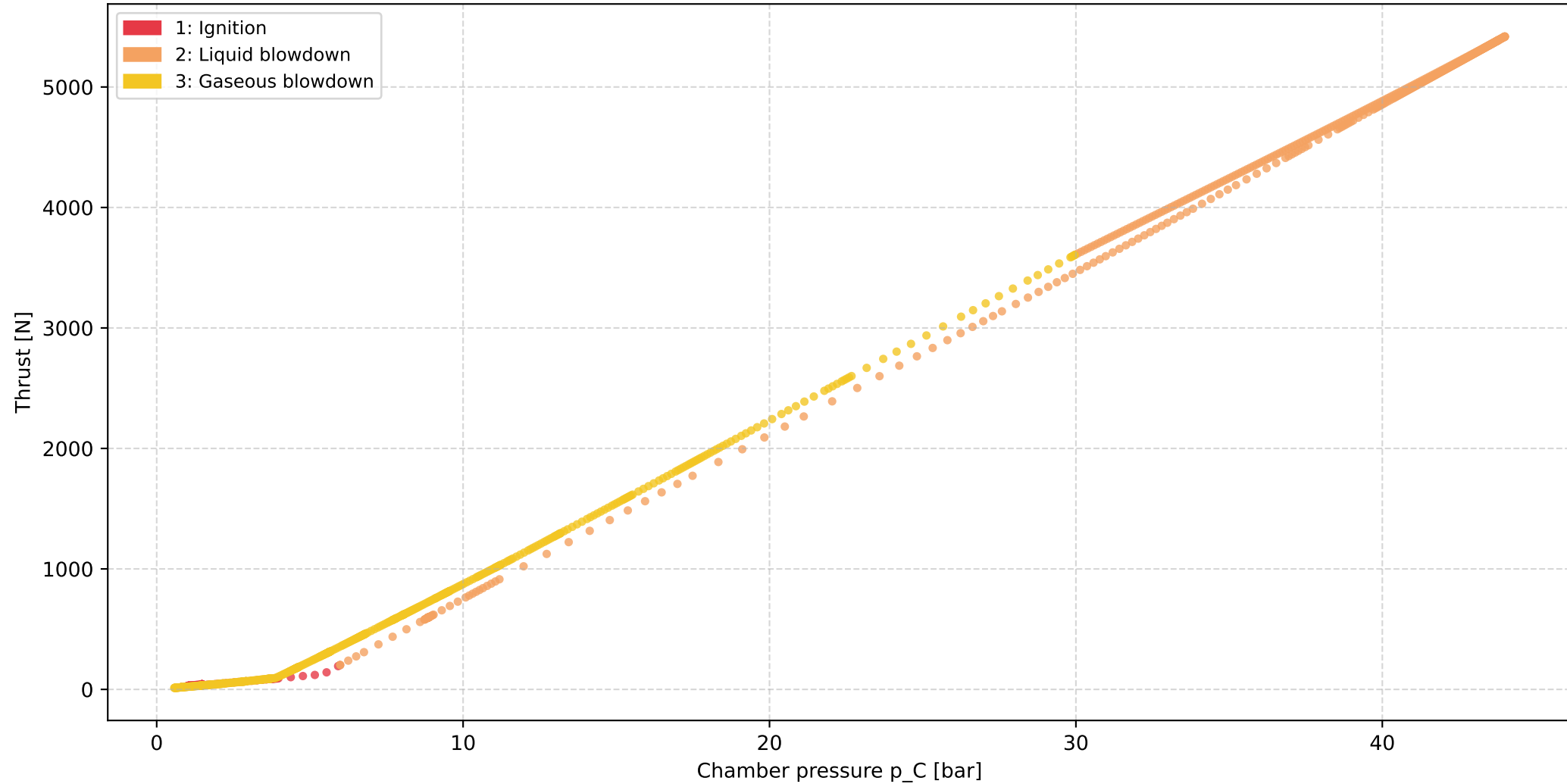
Trajectory — altitude vs. downrange



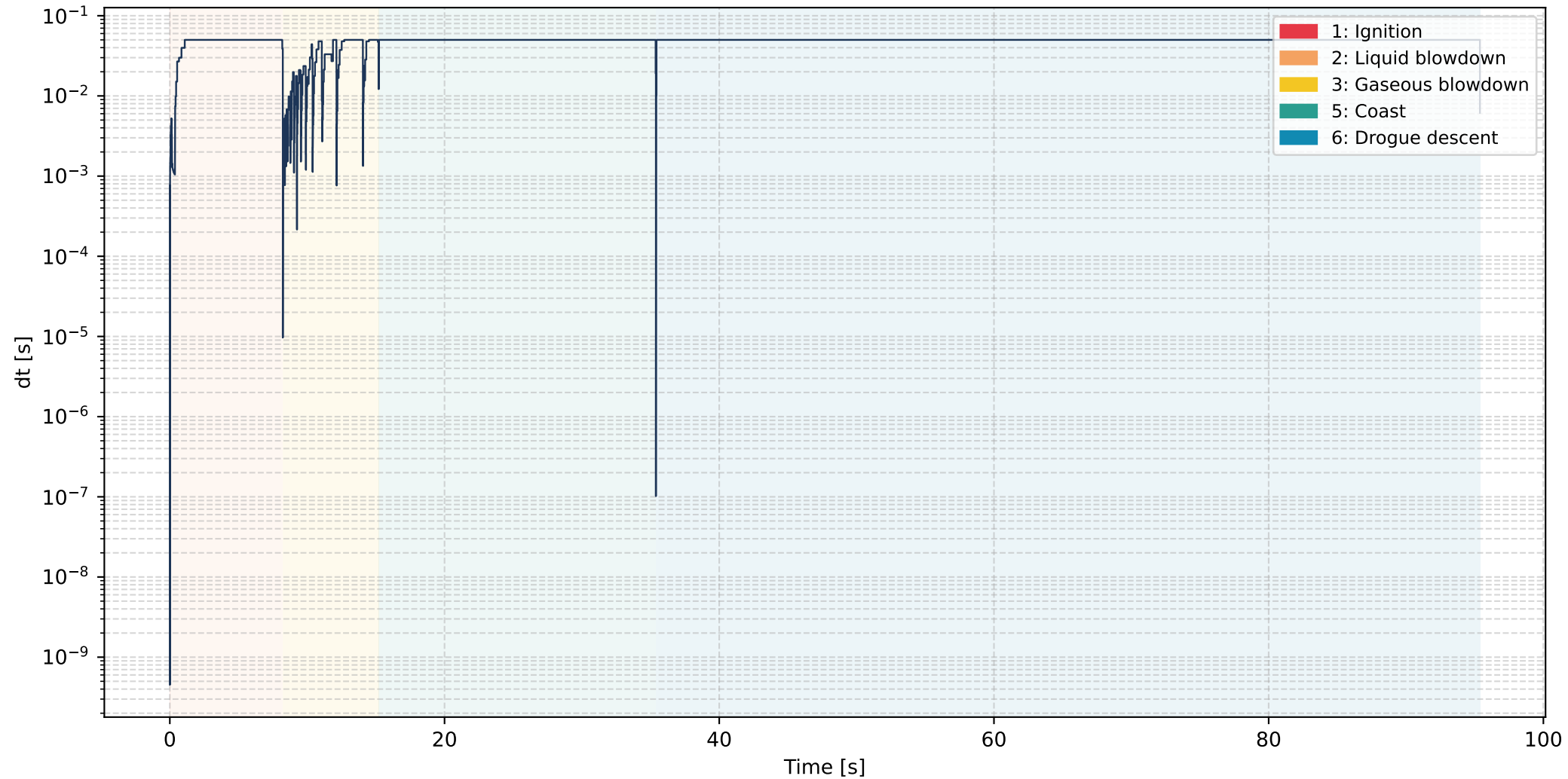
O/F ratio vs. fuel port radius (burn only)



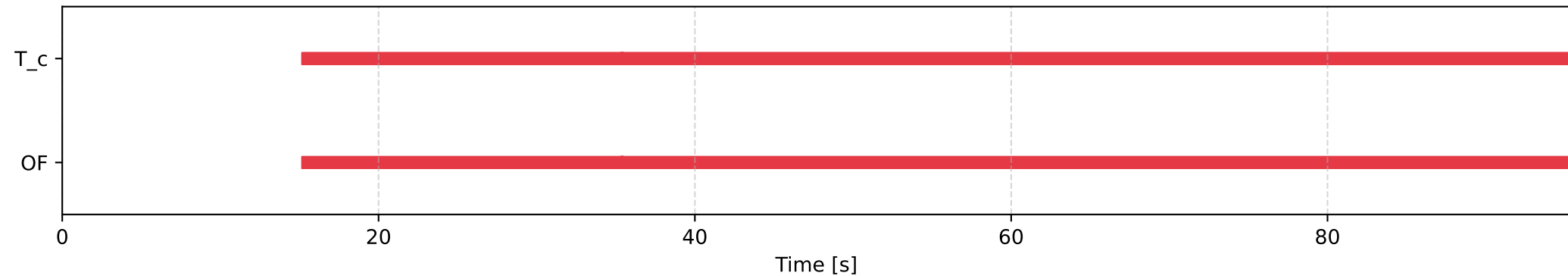
Thrust vs. chamber pressure (burn only)



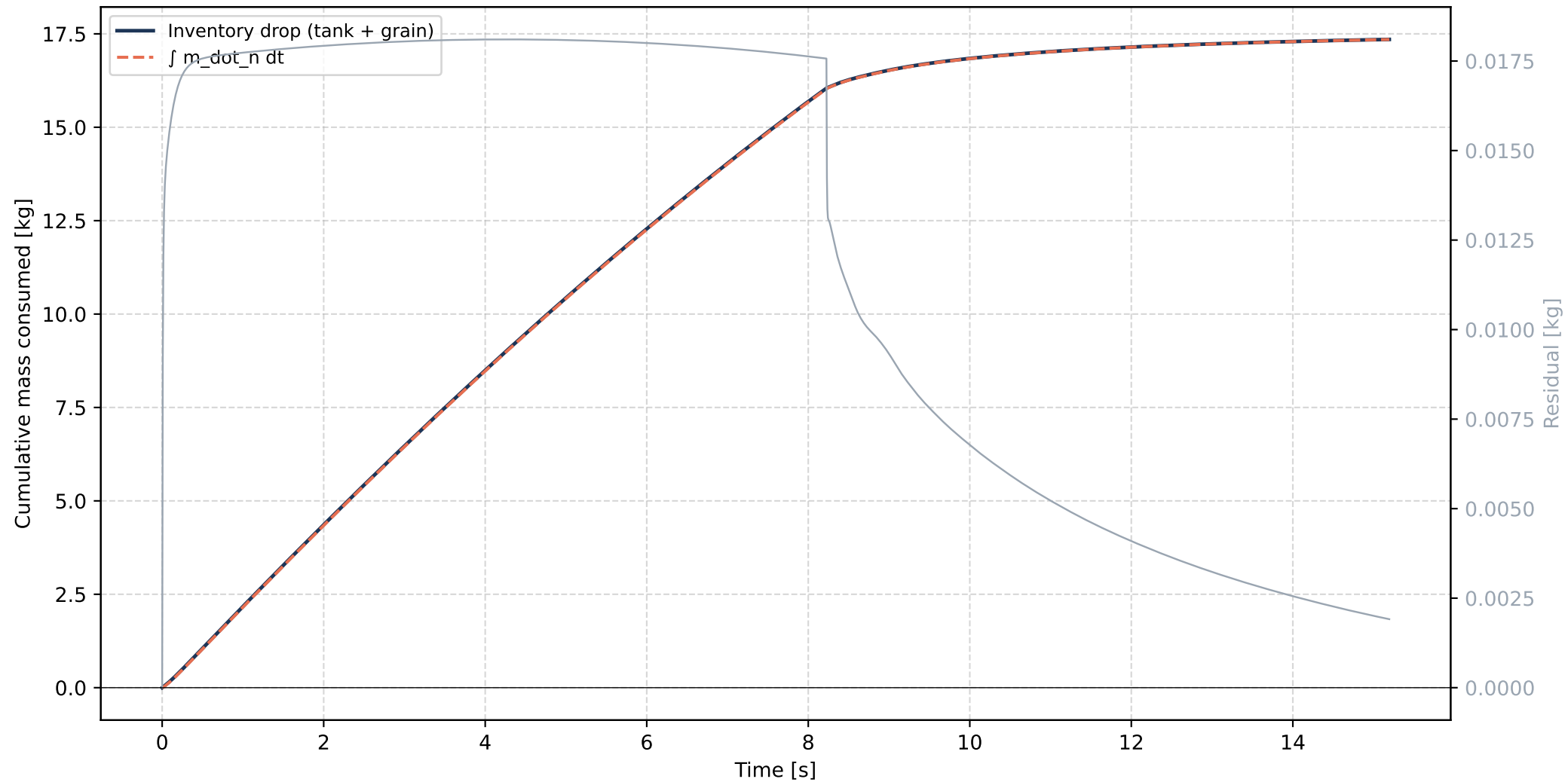
Solver step size dt vs. time



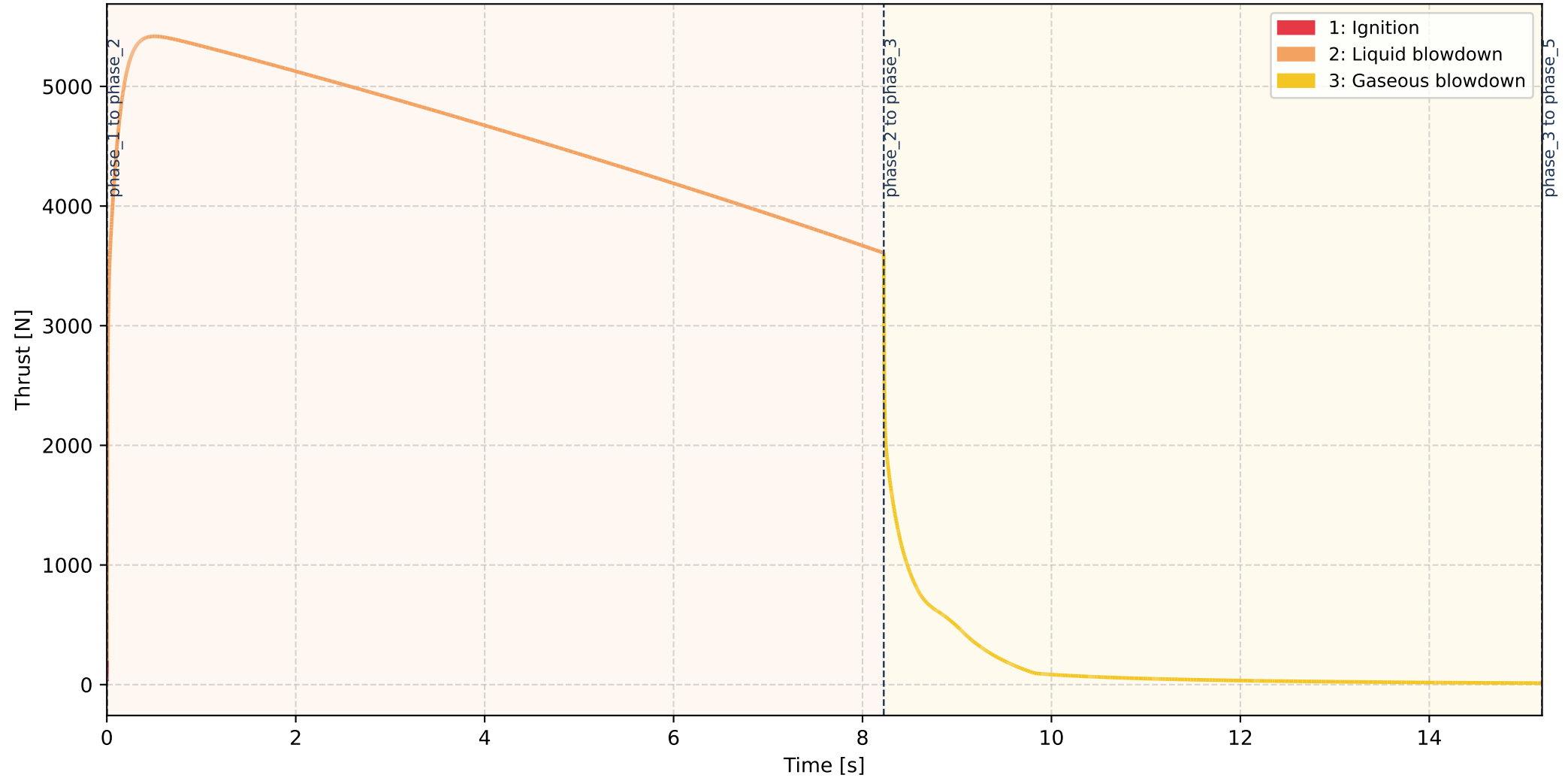
NaN map — variables with NaN values (2)



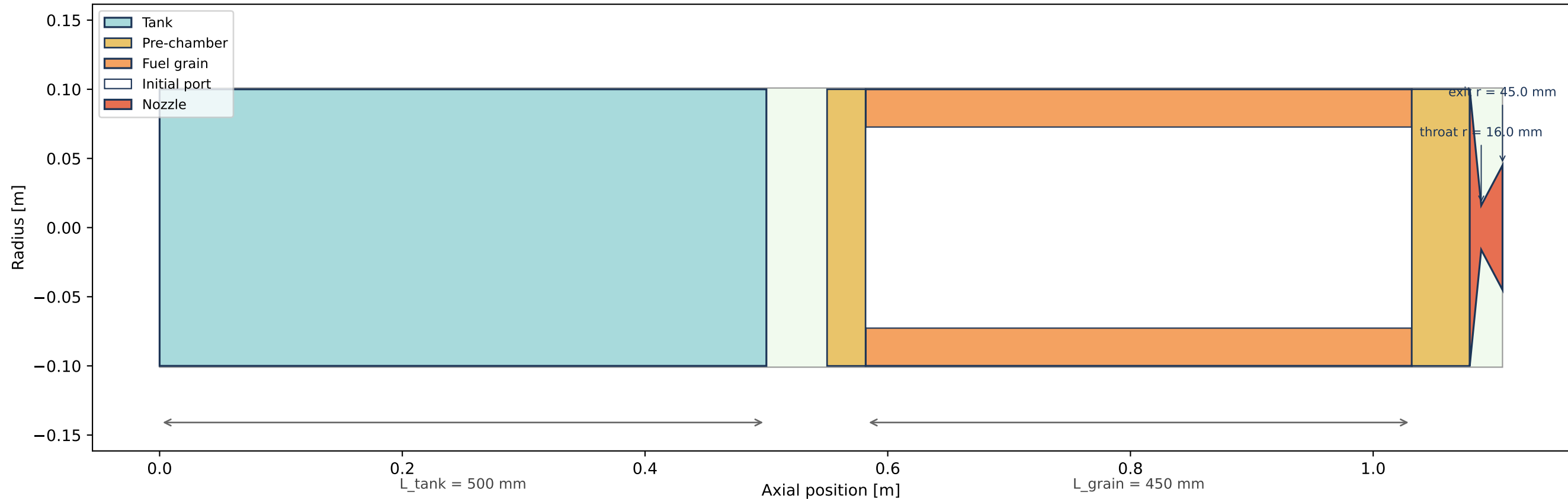
Mass conservation check (burn only)



Thrust with event markers (burn only)



Rocket cross-section (axisymmetric side view, axial-to-radial NOT to scale)



Nozzle profile — $\varepsilon = Ae/At = 7.91$

